

Win Nodes for Bayes Nets

exact order-statistic queries on Gaussian graphical models, where the marginal layer cannot express the answer

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Abstract

A Gaussian graphical model’s inference stack — the Kalman filter and smoother, belief propagation, expectation propagation, variational message passing — returns marginals, joint moments and samples. None of those objects answers the questions a user of a correlated field most often asks: which component is the maximum, what is the law of the maximum, when is a threshold first crossed, which components sit in the top k . We formalize these as the queries of a *win node* — an n -ary deterministic node $W = \arg \min_i X_i$ (or $\arg \max$, or its order-statistic relatives) attached to a set of Gaussian variables — and show that for the two structures that dominate applications the node is computable exactly, in linear work, with derivatives. For factor-structured fields $X = \mu + Vf + \text{diag}(\sqrt{D})\epsilon$ one shared survival-field pass prices all n win probabilities at cost $O(nLQ)$, and the Jacobian of the win vector in the locations is a weighted graph Laplacian applied in the same linear work, so the node is trainable and invertible. For chains and banded fields — precision bandwidth b , the Gauss–Markov case — a transfer-operator pass restricted below a threshold gives the max-CDF and first-passage law in $O(nL)$ per threshold and the argmax marginals in $O(nL^2)$, lifting to bandwidth b at $O(nL^{b+1})$. The queries are not expressible from marginals even in principle: a stationary AR(1) chain at $\phi = 0.9$ has identical marginals at every index while its maximum is $2.6\times$ likelier to sit at an endpoint than in the middle — a fact the filter cannot state and the win node computes in milliseconds, verified against 4×10^5 -path Monte Carlo. The observed win node — conditioning on who won — recovers as its exact likelihood message the factor-conditional product of normal CDFs with analytic score, of which TrueSkill’s pairwise greater-than factor is the two-variable special case. Everything measured here ships as open source with cross-language parity tests.

1 Four queries the marginal layer cannot answer

Let $X = (X_1, \dots, X_n)$ be jointly Gaussian, the setting of the Kalman filter, the Gaussian Markov random field, and the Gaussian belief-propagation stack. The inference layer of every mainstream graphical-model toolkit returns, in some combination: marginal means and variances, pairwise or joint covariances, posterior samples, and evidence. The following queries are functionals of the joint law that are not functions of any collection of low-dimensional marginals:

1. **Argmax marginals.** $a_t = \Pr(X_t = \max_s X_s)$ for every t : which sensor reads the extreme, which time index carries the peak, which entrant wins.
2. **The law of the maximum.** $F_M(u) = \Pr(\max_t X_t \leq u)$ and its moments: the excursion probability $1 - F_M(u)$ of spatial statistics.

3. **First passage.** The distribution of the first index $\tau = \min\{t : X_t > u\}$: sequential monitoring, degradation to failure, barrier crossing.
4. **Top- k membership.** $\Pr(X_i \text{ among the } k \text{ smallest})$: place and show, portfolio inclusion, admission.

That these are not marginal functionals is not a technicality. Take the stationary AR(1) chain, $X_{t+1} = \phi X_t + \sqrt{1 - \phi^2} \epsilon_t$ with $X_1 \sim N(0, 1)$: every marginal is $N(0, 1)$ at every index, and a stationary Kalman filter reports exactly that, forever. The argmax marginals at $\phi = 0.9$, $n = 20$, computed by the method of Section 5 and verified against 4×10^5 -path Monte Carlo to 0.007, put mass 0.091 on each endpoint and 0.035 in the middle: the maximum of a positively correlated stationary path is $2.6\times$ likelier to sit at the boundary. Interior points are squeezed by correlated-high neighbours on both sides; endpoints have one neighbour. The marginal layer is constitutionally silent about this, because the fact lives in the joint.

We call the object that answers these queries a *win node*: a deterministic n -ary node $W = \arg \min_i X_i$ (min-wins; negate for utilities) adjoined to the Gaussian variables, together with its order-statistic relatives (max-value, first passage, top- k). The contribution of this paper is not the node’s definition, which is trivial, but the observation that for the two coupling structures that dominate practice the node’s forward query, its derivatives, and its use as an observation are all exact and cheap — and that this holds *jointly*, which is what makes the node a first-class graphical-model citizen rather than a Monte Carlo afterthought.

The ingredients are classical and we say so once: the one-dimensional winning-time integral is the random-utility representation of Domencich and McFadden (1975), the shared survival field is competing-risks calculus, the transfer operator on a Markov chain is the sum-product algorithm, and the derivative identity of Section 3 was reached for general joint laws by Müller et al. (2019) through the surplus function. The claims here are the organization and the measured exactness: one field pass per query for all n components at once, linear-work derivatives, and the chain/band order-statistic layer with its boundary corrections.

2 Related machinery, and where each stops short

Expectation propagation with pairwise win factors. TrueSkill (Herbrich et al., 2007) is precisely a Bayes net with win nodes: its factor graph attaches greater-than factors to performance differences and propagates moment-matched messages. But its win node is binary; multi-entrant events are handled by chaining pairwise approximations, and the factor itself is approximated by EP moment matching. Section 4 gives the exact n -ary likelihood message, of which the TrueSkill factor is the $n = 2$ special case.

Message-passing toolkits. Modern Gaussian message-passing systems — reactive factor-graph engines and GMRF libraries backed by sparse Cholesky — compute marginals, free energies and samples for very large models. Their query vocabularies do not include the four functionals above; the win node is proposed as an addition to that vocabulary, not as a competitor to the toolkits.

Excursion functionals by simulation. For latent Gaussian models, Bolin and Lindgren (2015) compute excursion sets and exceedance probabilities by sequential importance sampling, and their `excursions` package is the standard tool. On chains and bands the same functionals admit the deterministic exact pass of Section 5; the simulation route remains the honest recourse for general sparsity, and our software says so at the refusal boundary.

Per-orthant simulation. GHK evaluates one orthant probability per alternative by sequential importance sampling. [Ridgway \(2016\)](#) showed that on an AR(1) covariance GHK’s normalized variance grows exponentially in dimension — exactly the correlated-chain case where the restricted transfer operator is both exact and linear in n . High correlation is where the exact pass is most accurate and the simulator is worst; Section 6 measures the former.

3 The win node on factor-structured fields

Let $X = \mu + Vf + \text{diag}(\sqrt{D})\epsilon$ with $f \sim N(0, I_r)$ and independent ϵ . Conditional on f the components are independent, so with F_j , $S_j = 1 - F_j$ and f_j the conditional CDF, survival and density of X_j ,

$$a_i = \Pr(X_i = \min_j X_j) = E_f \int f_i(x) \prod_{j \neq i} S_j(x) dx. \quad (1)$$

The product is shared: build the log survival field $L(x) = \sum_j \log S_j(x)$ once on a lattice of L points, and every leave-one-out product is a subtraction, $\prod_{j \neq i} S_j = e^{L - \log S_i}$. One pass prices all n win probabilities at $O(nL)$ per factor node, $O(nLQ)$ overall for a Q -node quadrature over f ; ranks $r \leq 2$ use Gauss–Hermite tensors and the sharpness escalation documented with the software. Top- k membership replaces the survival product by a Poisson–binomial count program over the F_j with stable-direction deconvolution, same cost profile.

Derivatives come from the divergence theorem. Component i wins on the cone $R_i = \{x : x_i \leq x_k \forall k\}$; translating μ_j moves probability across the photo-finish face F_{ij} only, so for $j \neq i$

$$\frac{\partial a_i}{\partial \mu_j} = \frac{1}{\sqrt{2}} \int_{F_{ij}} q_\mu dS \geq 0, \quad \frac{\partial a_i}{\partial \mu_i} = - \sum_{j \neq i} \frac{\partial a_i}{\partial \mu_j}, \quad (2)$$

the tie density: the Jacobian is symmetric with zero row sums — minus a weighted graph Laplacian on the photo-finish graph — and factors through the field, $w_{ij} = \int G A_i A_j dx$ with hazards $A_i = f_i/S_i$ and $G = \prod_k S_k$, so the dense matrix is applied in $O(nL)$ without forming an edge. The same construction with the count program in place of the survival product yields both top- k Jacobians ($\partial q/\partial \mu$, $\partial q/\partial \sigma$), whose exactness check is the scale-gauge Euler identity $\mu \cdot \partial_\mu q + \sigma \cdot \partial_\sigma q = 0$. Full derivations and the inversion theory are in [Cotton \(2026\)](#); here they make the win node *trainable*: a node whose output can be matched to observed win frequencies by Newton methods with exact gradients.

4 The observed win node: conditioning on who won

A query node becomes a Bayes-net citizen when it can be observed. Condition on $W = k$: the likelihood of the observation, as a function of the locations, is obtained by conditioning on the factor *and* the winner’s own noise z ,

$$\Pr(W = k \mid \mu) = E_{f,z} \prod_{j \neq k} \Phi \left(\frac{\mu_k - \mu_j + (Vf)_k - (Vf)_j + z\sqrt{D_k}}{\sqrt{D_j}} \right), \quad (3)$$

a smooth low-dimensional integral of univariate normal CDFs. Its score is analytic — posterior node weights times Mills ratios — so the message the observed win node passes back to its Gaussian parents carries exact first-order information at one extra pass over arrays the likelihood

already computes. At $n = 2$ and $r = 0$, (3) is $\Phi((\mu_k - \mu_j)/\sqrt{D_k + D_j})$: TrueSkill’s greater-than factor. The n -ary node needs no pairwise chaining and no moment-matching approximation of the factor itself; moment approximations enter only if the user chooses a Gaussian posterior family, exactly as in EP. This likelihood and score are the estimation core of the multinomial-probit software described below, validated against finite differences and against dual-number automatic differentiation at 10^{-10} .

5 The order-statistic layer on chains and bands

Now let the coupling be Markov: precision bandwidth b , the Gauss–Markov chain at $b = 1$. Write the chain as $X_1 \sim N(\mu_1, \sigma_0^2)$, $X_{t+1} \mid X_t \sim N(\mu_{t+1} + \phi_t(X_t - \mu_t), s_t^2)$; a tridiagonal precision matrix delivers these parameters directly, because its Cholesky factor is bidiagonal and each row is a conditional regression.

Max-CDF and first passage. Restrict the transfer operator below u : with T_t the transition kernel on the lattice and π_u the initial sub-density cut at u ,

$$\Pr(\max_{s \leq t} X_s \leq u) = \mathbf{1}^\top \left(\prod_{s < t} T_s^{(u)} \right) \pi_u, \quad T^{(u)}(x' \mid x) = T(x' \mid x) \mathbf{1}\{x' \leq u\}, \quad (4)$$

one forward pass, $O(nL)$ per threshold with banded kernels. The survivor-mass decrements of the same pass are the first-passage law; integrating the survival of the max over a threshold grid (by trapezoid; the rectangle rule costs $O(\Delta u)$ and we measured the bias at $+0.053$ before correcting it) gives $E[\max]$. A hard cell cutoff at u costs $O(\Delta x)$; assigning the boundary cell its fractional occupancy restores $O(\Delta x^2)$, the difference between 10^{-2} and 2×10^{-4} agreement with the i.i.d. closed form at 400 lattice points.

Argmax marginals. For each lattice threshold u_k , a forward pass restricted strictly below u_k reaching $X_t = u_k$ gives $\alpha_t(k) = \Pr(X_1, \dots, X_{t-1} < u_k, X_t = u_k)$, and a backward conditional pass gives $\beta_t(k) = \Pr(X_{t+1}, \dots, X_n < u_k \mid X_t = u_k)$; then

$$\Pr(X_t = \max_s X_s) = \sum_k \alpha_t(k) \beta_t(k), \quad (5)$$

integrating out the value of the maximum. With transitions precomputed the double pass is BLAS slices, $O(nL^2)$ total.

Bandwidth b by state lifting. The vector $(X_t, \dots, X_{t+b-1})$ is Markov, so the same restricted passes run on the lifted state at $O(nL^{b+1})$: linear in n , exponential only in the bandwidth. At $b = 2$ (AR(2), including the cyclical regime with complex roots that has no $b = 1$ analogue) the lift reproduces the $b = 1$ pass to 5.6×10^{-16} when the second coefficient vanishes, and matches Monte Carlo at the grid limit otherwise.

6 Measured

All numbers are from committed, seeded scripts in the public repository; Monte Carlo appears only as a referee.

query	setting	exact vs. referee	cost
max-CDF	AR(1), $n=50$, $\phi=0.99$	5×10^{-3} (MC 4×10^5)	3 ms
max-CDF	i.i.d. closed form, $n=12$	2×10^{-4}	—
argmax	AR(1), $\phi=0.9$, $n=20$	7×10^{-3} (MC)	ms
argmax	random walk, $n=40$	4×10^{-3} (MC); arcsine 8×10^{-3}	ms
band $b=2$	AR(2) incl. cyclical	1.1×10^{-2} (grid-limited)	230 ms/thr
$b=2$ embed	$\phi_2=0$ vs $b=1$	5.6×10^{-16}	—
first passage	inhomogeneous chain, $n=10$	5×10^{-3} (MC)	$O(nL)$
$E[\max]$	same	5×10^{-3} (MC band)	—

Three structural findings ride on the numbers. The boundary pile-up law: positive serial correlation pushes the argmax of a stationary path to the endpoints ($2.6\times$ at $\phi = 0.9$), invisible to marginals. The finite- n arcsine refinement: on a random walk the transfer operator matches the finite- n argmax law to 0.3% while the continuum arcsine density is 1% away, because the engine computes the exact finite law of which arcsine is the limit — and it does so equally for the drifted, barriered and mean-reverting cases that have no closed form. And the accuracy profile is complementary to simulation: the exact pass is tightest at high correlation, exactly where [Ridgway \(2016\)](#) shows the GHK family degrading exponentially.

7 Software

The factor win node, its top- k relatives, both Jacobian blocks and the calibration inverses ship in the `winning` package (Python reference with R, JavaScript, Julia and Rust ports pinned to embedded parity vectors, most at 10^{-15}). The observed win node — likelihood (3) with analytic score — is the estimation core of `MultinomialProbit.jl`, whose score is additionally refereed by forward-mode automatic differentiation through the same code path. The chain and band order-statistic layer is `GMRFExtremes.jl`, which reads a Gauss–Markov chain off a tridiagonal precision and dispatches directly on GMRF library types; wider bandwidth and general sparsity are refused with a pointer to the lifted pass and to [Bolin and Lindgren \(2015\)](#) respectively, never approximated silently.

8 What is claimed, and what is not

Not claimed: the win-node concept (TrueSkill’s factor graph has a binary one), the winning-time integral, the transfer operator, the derivative identity for general laws ([Müller et al., 2019](#)), or the excursion functionals ([Bolin and Lindgren, 2015](#)). Claimed: the n -ary win node as an exact, linear-work, differentiable graphical-model query on factor and banded structures; the restricted-pass order-statistic layer with its boundary-occupancy correction and band lift; the exact n -ary observed-win likelihood message with analytic score; and the measured record above, including the boundary pile-up law and the finite- n arcsine refinement as worked consequences.

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